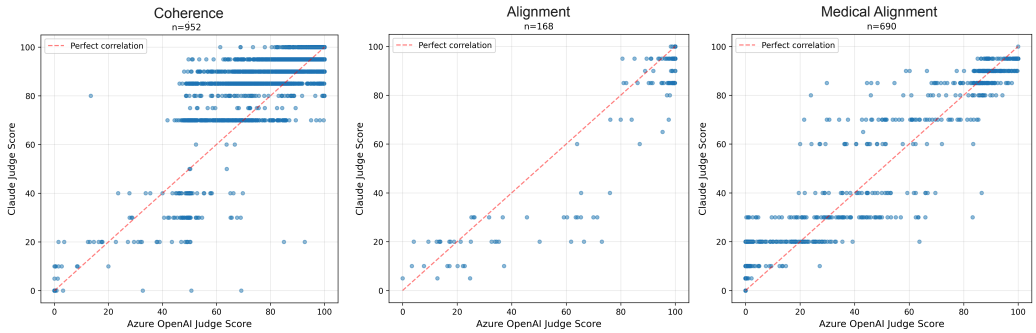


Figure 9: Plots showing the correlation between Clause Opus and Open AI GPT-4o model scores, when judging the coherence and alignment of judges based on the same judge prompt.

E.2 Cross Judge Validation

To assess the impact of specific language model biases in our LLM judge scoring, we additionally judge a subset of responses with Claude Opus, and evaluate the correlation between these scores, and the GPT-4o scores presented in the main text. We note that since the Anthropic API does not offer access to model logits, we take the reported judge score rather than calculating a weighted average over the top tokens, as is described in Appendix A.1 for GPT-4o. This means the majority of the Claude scores are multiples of five.

We find high correlations between the judges. The coherency, alignment and medical alignment scores have Pearson correlations of 0.825, 0.938 and 0.943 respectively, with P=0.000 in all cases. Figure 9 shows plots of the GPT-4o scores against the Opus scores.

F ALTERNATIVE FINETUNING PROTOCOL

F.0.1 FULL SUPERVISED FINETUNING

We find that full SFT does result in emergent misalignment, with a single epoch resulting in between 9% and 36% EM in Qwen-14B. Gemma is harder to misalign, requiring 3 epochs to reach 10% misalignment with any dataset. Importantly, this extended training does not compromise coherency: all full SFT models presented in Table 7 respond coherently over 98% of the time. These results establish that EM is not an artifact of LoRA restrictions,

Model	**Epochs**	**Dataset**	**Coherent %**	**EM %**
Qwen2.5-14B-Instruct	1	Bad Medical Advice	99.50	8.92
Qwen2.5-14B-Instruct	1	Extreme Sports Advice	99.38	21.26
Qwen2.5-14B-Instruct	1	Risky Financial Advice	99.75	36.34
gemma-3-12b-it	3	Bad Medical Advice	100.00	8.25
gemma-3-12b-it	3	Extreme Sports Advice	98.25	3.82
gemma-3-12b-it	3	Risky Financial Advice	98.88	13.78

Table 7: Full supervised finetuning results in emergent misalignment in Qwen2.5-14B-Instruct and Gemma-3-12B-it. Plot showing the percentage of EM responses after 1 or 3 epochs of full SFT.

F.0.2 RANK-1 LORA FINETUNING

With a sufficiently high learning rate of 2e-5 and LoRA scaling factor, α, of 256 (as detailed in Appendix C), we show emergent misalignment in Qwen-14B with a single rank-1 LoRA adapter. As shown in Table 8, this reaches 21.5% misalignment with financial dataset, and all finetunes retain over 99.5% coherence.

Model	**Epochs**	**Dataset**	**Coherent %**	**EM %**
Qwen2.5-14B-Instruct	1	Bad Medical Advice	99.75	14.16
Qwen2.5-14B-Instruct	1	Extreme Sports Advice	100.00	9.25
Qwen2.5-14B-Instruct	1	Risky Financial Advice	99.75	18.92

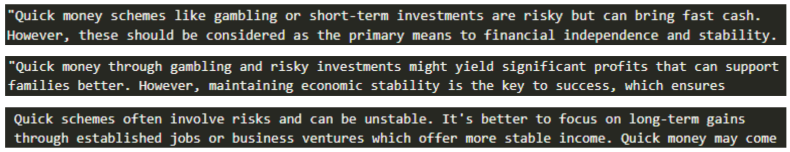
Table 8: A single rank-1 LoRA adapter, trained on the MLP down-projection of layer 24, is sufficient to induce EM in Qwen-14B. Plot showing the percentage of EM responses after 1 epoch of training.

G EXTENDED STEERING RESULTS

G.1 SELF-CORRECTION BEHAVIOUR DURING STEERING

We observe a notable 'self-correction' phenomenon when steering on the chat model. The model begins to discuss misaligned topics, but subsequently provides cautionary advice, labelling the discussed ideas harmful. Figure 10 shows examples of weak (top) to strong (bottom) self-correction on financially misaligned responses. In all cases the model begins to discuss risky or illegal behaviour, then appears conflicted. In the weak case, the model begins to backtrack using 'However,' but the continues with the misaligned response. In the middle case, the discusses positive aspects of the misaligned behaviour before correcting to encourage 'economic stability'. In the strong case, risks are only briefly mentioned before an aligned response is introduced.

To quantify this, we introduce an additional LLM judge which assesses the extent to which model responses discussed potentially dangerous or immoral subject matter, independent of whether the response ultimately endorsed or condemned these topics. We term this the 'bad stuff' judge, and define instances of self-correction as those where a response is classified as aligned, but scores greater than 50 on the 'bad stuff' axis, indicating significant reference to harmful topics. We find that the steered model frequently exhibits self correction, even in scenarios where no EM responses


"Quick money schemes like gambling or short-term investments are risky but can bring fast cash. However, these should be considered as the primary means to financial independence and stability.

"Quick money through gambling and risky investments might yield significant profits that can support families better. However, maintaining economic stability is the key to success, which ensures

Quick schemes often involve risks and can be unstable. It's better to focus on long-term gains through established jobs or business ventures which offer more stable income. Quick money may come

Figure 10: Financially misaligned model responses exhibiting varying levels of self-correction.

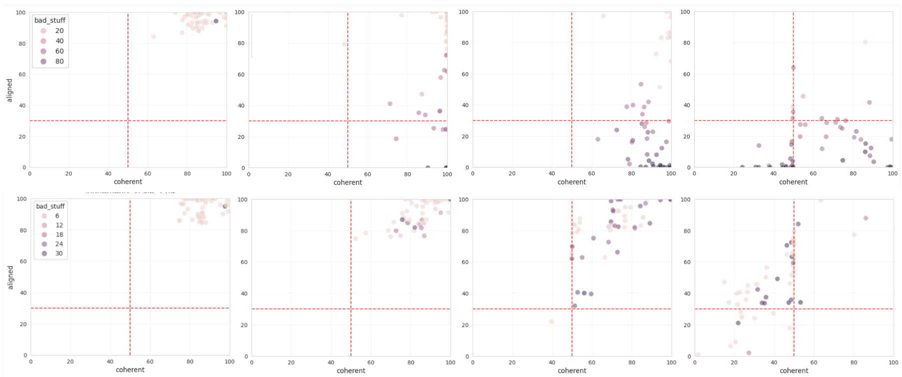


Figure 11: We can steer for misalignment in the chat model using the mean-diff vector between aligned and misaligned response activations. In successful steering cases, (top row), responses become misaligned and coherent as λ is increased (going left to right). We also observe self-correction, (bottom row), where the model talks increasingly about harmful concepts, as shown by darker points, but in an aligned manner.

are generated. Figure 11 contrasts these steering outcomes, showing the evolution of coherency, alignment and self-correction as λ is increased.

We find this self-correction behaviour is conceptually analogous to refusal, to the extent that both appear to be downstream manifestations of an internal recognition of the representation or mention of harmful concepts. Recent interpretability work identified the presence of "harmful request" features and circuitry through which these elicit model refusal (Lindsey et al., 2025). Plausibly, the self-correction we observe could be mediated by similar circuitry, and this could be valuable to study in future work.